

Formal Logic (SM234614)

Introduction

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1. Bio of Lecturer
2. Learning Outcomes
3. Plans of Study
4. Previous Projects
5. Rules



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The students are

- ▶ able to understand the basics of Boolean logic and their operations



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- ▶ able to understand the **Boolean Satisfiability**



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- ▶ able to model and to solve mathematical and computational problems using Boolean satisfiability



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- ▶ able to model and to solve mathematical and computational problems using Boolean satisfiability
- ▶ able to understand the **Satisfiability Modulo Theory**



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- ▶ able to model and to solve mathematical and computational problems using Boolean satisfiability
- ▶ able to understand the **Satisfiability Modulo Theory**
- ▶ able to model and to solve mathematical and computational problems using Satisfiability Modulo Theory



Weeks 1-2

- ▶ Students understand the basics of Boolean logic and their operations
- ▶ Students understand the CNF dan DNF encodings
- ▶ Students understand the Karnaugh Mapping



Weeks 1-2

- ▶ Students understand the basics of Boolean logic and their operations
- ▶ Students understand the CNF dan DNF encodings
- ▶ Students understand the Karnaugh Mapping

Weeks 3-5

- ▶ Students understand the Boolean Satisfiability
- ▶ Students are able to apply algorithms to solve Boolean Satisfiability problems



Weeks 6-7

- ▶ Students are able to model and to solve mathematical and computational problems using Boolean Satisfiability



Weeks 6-7

- ▶ Students are able to model and to solve mathematical and computational problems using Boolean Satisfiability

Week 8

- ▶ Midterm Exam



Weeks 9-11

- ▶ Students understand the Satisfiability Modulo Theory



Weeks 9-11

- ▶ Students understand the Satisfiability Modulo Theory

Weeks 12-15

- ▶ Students are able to model and to solve mathematical and computational problems using Satisfiability Modulo Theory



Weeks 9-11

- ▶ Students understand the Satisfiability Modulo Theory

Weeks 12-15

- ▶ Students are able to model and to solve mathematical and computational problems using Satisfiability Modulo Theory

Week 16

- ▶ Final Exam



Project #1

Find A, B, C, D, E such that $ABCD \times E = DCBA, E > 1$



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Project #2

Determine whether $f(x) = 2x^4 + 3x^3 + x^2 + 4x + 1$ is reducible in $\mathbb{Z}_5[x]$.
If so, find all polynomials $p(x)$ and $q(x)$ such that $p(x) \times q(x) = f(x)$.



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If so, find all polynomials $p(x)$ and $q(x)$ such that $p(x) \times q(x) = f(x)$.

Project #3

Determine the number of ways to color $1 \times n$ tiles with three different colors (red, blue, green) with the following rules: (i) the number of red tiles are even; (ii) the number of blue tiles is at least one.



Project #4

There are five houses, each with a different color. In each house lives a person from a different city in Indonesia. Each homeowner has a favorite food, a specific hobby, and a particular pet. No two homeowners have the same pet, the same favorite food, or the same hobby.

Additional information:



Project #4

There are five houses, each with a different color. In each house lives a person from a different city in Indonesia. Each homeowner has a favorite food, a specific hobby, and a particular pet. No two homeowners have the same pet, the same favorite food, or the same hobby.

Additional information:

- ▶ The person from Jakarta lives in the red house.



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Additional information:

- ▶ The person from Jakarta lives in the red house.
- ▶ The person from Surabaya has a dog.



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Additional information:

- ▶ The person from Jakarta lives in the red house.
- ▶ The person from Surabaya has a dog.
- ▶ The person from Bandung likes to eat fried rice.



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Additional information:

- ▶ The person from Jakarta lives in the red house.
- ▶ The person from Surabaya has a dog.
- ▶ The person from Bandung likes to eat fried rice.
- ▶ The green house is immediately to the left of the white house.



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Additional information:

- ▶ The person from Jakarta lives in the red house.
- ▶ The person from Surabaya has a dog.
- ▶ The person from Bandung likes to eat fried rice.
- ▶ The green house is immediately to the left of the white house.
- ▶ The owner of the green house likes to eat satay.



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Additional information:

- ▶ The person from Jakarta lives in the red house.
- ▶ The person from Surabaya has a dog.
- ▶ The person from Bandung likes to eat fried rice.
- ▶ The green house is immediately to the left of the white house.
- ▶ The owner of the green house likes to eat satay.
- ▶ The person whose hobby is reading keeps a bird.



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- ▶ The owner of the green house likes to eat satay.
- ▶ The person whose hobby is reading keeps a bird.
- ▶ The owner of the yellow house enjoys gardening.



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- ▶ The person from Surabaya has a dog.
- ▶ The person from Bandung likes to eat fried rice.
- ▶ The green house is immediately to the left of the white house.
- ▶ The owner of the green house likes to eat satay.
- ▶ The person whose hobby is reading keeps a bird.
- ▶ The owner of the yellow house enjoys gardening.
- ▶ The person living in the middle house likes to eat meatballs.



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Additional information:

- ▶ The person from Medan lives in the first house.



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Additional information:

- ▶ The person from Medan lives in the first house.
- ▶ The person whose hobby is cooking lives next to the person who keeps a cat.



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Additional information:

- ▶ The person from Medan lives in the first house.
- ▶ The person whose hobby is cooking lives next to the person who keeps a cat.
- ▶ The person who keeps a horse lives next to the person who enjoys gardening.



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Additional information:

- ▶ The person from Medan lives in the first house.
- ▶ The person whose hobby is cooking lives next to the person who keeps a cat.
- ▶ The person who keeps a horse lives next to the person who enjoys gardening.
- ▶ The person whose hobby is playing music likes to eat rendang.



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Additional information:

- ▶ The person from Medan lives in the first house.
- ▶ The person whose hobby is cooking lives next to the person who keeps a cat.
- ▶ The person who keeps a horse lives next to the person who enjoys gardening.
- ▶ The person whose hobby is playing music likes to eat rendang.
- ▶ The person from Yogyakarta enjoys painting.



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- ▶ The person whose hobby is cooking lives next to the person who keeps a cat.
- ▶ The person who keeps a horse lives next to the person who enjoys gardening.
- ▶ The person whose hobby is playing music likes to eat rendang.
- ▶ The person from Yogyakarta enjoys painting.
- ▶ The person from Medan lives next to the blue house.



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- ▶ The person whose hobby is playing music likes to eat rendang.
- ▶ The person from Yogyakarta enjoys painting.
- ▶ The person from Medan lives next to the blue house.
- ▶ The person whose hobby is cooking has a neighbor who likes to eat gado-gado.



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- ▶ The person from Yogyakarta enjoys painting.
- ▶ The person from Medan lives next to the blue house.
- ▶ The person whose hobby is cooking has a neighbor who likes to eat gado-gado.

Determine all possible arrangements!



- ▶ Student-centered learning
- ▶ Tolerance for coming late: 10 minutes
- ▶ No cheating in any form
- ▶ Drinks are allowed
- ▶ Snacks are not allowed
- ▶